

Report II

Advanced Statistical Mechanics / Statistical Mechanics I

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The final results can be found in (1.4), (2.3), (3.4), (4.1), (5.7), (6.2), (7.5) and (8.3).

1 Gaussian integral

Let $\mathbf{x} = (x_1, \dots, x_N)$ be an N -dimensional real vector, and let A be an $N \times N$ real, symmetric, and positive-definite matrix.

(1) Calculate the Gaussian integral

$$\int_{-\infty}^{\infty} d^N x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}} \left(= \int_{-\infty}^{\infty} dx_1 \cdots \int_{-\infty}^{\infty} dx_N e^{-\frac{1}{2} \sum_{i,j=1}^N x_i A_{ij} x_j} \right). \quad (1)$$

Solution. Since A is a real symmetric $N \times N$ matrix, we can diagonalize it using an orthogonal matrix,

$$P^T A P = \text{diag}(a_1, a_2, \dots, a_N), \quad P \in SO(N) \quad (1.1)$$

Using the substitution $\mathbf{x} \rightarrow P \cdot \mathbf{x}$, we obtain

$$\int_{-\infty}^{\infty} d^N x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}} = \frac{1}{\det P} \int_{-\infty}^{\infty} dx_1 \cdots \int_{-\infty}^{\infty} dx_N e^{-\frac{1}{2} \sum_{i=1}^N a_i x_i^2} = \prod_{i=1}^N \int_{-\infty}^{\infty} dx_i e^{-\frac{1}{2} a_i x_i^2} \quad (1.2)$$

Using the following formula,

$$\int_{-\infty}^{\infty} dx e^{-ax^2} = \sqrt{\frac{\pi}{a}} \quad (1.3)$$

We obtain the final answer:

Final Answer

$$\prod_{i=1}^N \sqrt{\frac{2\pi}{a_i}} = \sqrt{\frac{(2\pi)^N}{\det A}} \quad (1.4)$$

(2) For an N -dimensional real vector $\mathbf{b} = (b_1, \dots, b_N)$, calculate the Gaussian integral

$$\int_{-\infty}^{\infty} d^N x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x}}. \quad (2)$$

Solution. Diagonalizing A as in the previous problem and denoting $P^T \cdot \mathbf{b}$ by $(b'_1, \dots, b'_N)$, we have

$$\begin{aligned} \int_{-\infty}^{\infty} d^N x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x}} &= \int_{-\infty}^{\infty} dx_1 \cdots \int_{-\infty}^{\infty} dx_N e^{-\frac{1}{2} \sum_{i=1}^N a_i x_i^2 + b'_i x_i} \\ &= \prod_{i=1}^N \int_{-\infty}^{\infty} dx_i e^{-\frac{1}{2} a_i x_i^2 + b'_i x_i} = \prod_{i=1}^N \int_{-\infty}^{\infty} dx_i e^{-\frac{1}{2} a_i (x_i - \frac{b'_i}{a_i})^2} \cdot e^{\frac{b_i'^2}{2a_i}} \\ &= \sqrt{\frac{(2\pi)^N}{\det A}} \cdot e^{\sum_{i=1}^N \frac{b_i'^2}{2a_i}} \end{aligned} \quad (2.1)$$

Notice that

$$\sum_{i=1}^N \frac{b_i'^2}{2a_i} = \frac{1}{2} (P^T \cdot \mathbf{b})^T \cdot (P^T A P)^{-1} \cdot P^T \cdot \mathbf{b} = \frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b} \quad (2.2)$$

Then the final answer is:

Final Answer

$$\sqrt{\frac{(2\pi)^N}{\det A}} \cdot e^{\frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b}} \quad (2.3)$$

For a function $f(\mathbf{x})$ of the N -dimensional vector $\mathbf{x}$, we introduce the average over the Gaussian distribution:

$$\langle f(\mathbf{x}) \rangle := \frac{\int_{-\infty}^{\infty} d^N x f(\mathbf{x}) e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}}}{\int_{-\infty}^{\infty} d^N x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}}} \quad (3)$$

(3) Calculate $\langle x_i x_j \rangle$ ($i, j \in \{1, \dots, N\}$).

Solution. With the following notation:

$$Z[\mathbf{b}] := \int_{-\infty}^{\infty} d^N x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x}} \quad (3.1)$$

It is easy to show that

$$\begin{aligned} \langle x_i \rangle &:= \frac{\int_{-\infty}^{\infty} d^N x x_i e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}}}{\int_{-\infty}^{\infty} d^N x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}}} = \frac{1}{Z[0]} \frac{\partial}{\partial b_i} Z[\mathbf{b}] \Big|_{\mathbf{b}=0} \\ \langle x_i x_j \rangle &:= \frac{\int_{-\infty}^{\infty} d^N x x_i x_j e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}}}{\int_{-\infty}^{\infty} d^N x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}}} = \frac{1}{Z[0]} \frac{\partial}{\partial b_i} \frac{\partial}{\partial b_j} Z[\mathbf{b}] \Big|_{\mathbf{b}=0} \\ &\dots \\ \langle x_{i_1} \cdots x_{i_n} \rangle &:= \frac{\int_{-\infty}^{\infty} d^N x x_{i_1} \cdots x_{i_n} e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}}}{\int_{-\infty}^{\infty} d^N x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x}}} = \frac{1}{Z[0]} \frac{\partial}{\partial b_{i_1}} \cdots \frac{\partial}{\partial b_{i_n}} Z[\mathbf{b}] \Big|_{\mathbf{b}=0} \end{aligned} \quad (3.2)$$

Focusing on $\langle x_i x_j \rangle$ and using (2.3), we obtain:

$$\begin{aligned}
 \langle x_i x_j \rangle &= \frac{1}{Z[0]} \frac{\partial}{\partial b_i} \frac{\partial}{\partial b_j} Z[\mathbf{b}] \Big|_{\mathbf{b}=0} = \sqrt{\frac{(2\pi)^N}{\det A}} \frac{1}{Z[0]} \frac{\partial}{\partial b_i} \frac{\partial}{\partial b_j} e^{\frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b}} \Big|_{\mathbf{b}=0} \\
 &= e^{\frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b}} \frac{\partial}{\partial b_i} \frac{\partial}{\partial b_j} \left(\frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b} \right) \Big|_{\mathbf{b}=0} + \left[\frac{\partial}{\partial b_i} e^{\frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b}} \frac{\partial}{\partial b_j} \left(\frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b} \right) \Big|_{\mathbf{b}=0} + (i \leftrightarrow j) \right] \\
 &= \frac{1}{2} \left((A^{-1})_{ij} + (A^{-1})_{ji} \right)
 \end{aligned} \tag{3.3}$$

Since A is symmetric, so is A^{-1} . Therefore, the final answer is:

Final Answer

$$\langle x_i x_j \rangle = (A^{-1})_{ij} \tag{3.4}$$

(4) Show

$$\langle x_i x_j x_k x_l \rangle = \langle x_i x_j \rangle \langle x_k x_l \rangle + \langle x_i x_k \rangle \langle x_j x_l \rangle + \langle x_i x_l \rangle \langle x_j x_k \rangle \quad (i, j, k, l \in \{1, \dots, N\}). \tag{4}$$

Solution. The proof is straightforward by definition. Notice that the only contribution comes from the second-order term in the Taylor expansion of $e^{\frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b}}$, which is the fourth-order term in $\mathbf{b}$.

Final Answer

$$\begin{aligned}
 \langle x_i x_j x_k x_l \rangle &= \frac{1}{Z[0]} \frac{\partial}{\partial b_i} \frac{\partial}{\partial b_j} \frac{\partial}{\partial b_k} \frac{\partial}{\partial b_l} Z[\mathbf{b}] \Big|_{\mathbf{b}=0} \\
 &= \sqrt{\frac{(2\pi)^N}{\det A}} \frac{1}{Z[0]} \frac{\partial}{\partial b_i} \frac{\partial}{\partial b_j} \frac{\partial}{\partial b_k} \frac{\partial}{\partial b_l} e^{\frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b}} \Big|_{\mathbf{b}=0} \\
 &= \frac{\partial}{\partial b_i} \frac{\partial}{\partial b_j} \frac{\partial}{\partial b_k} \frac{\partial}{\partial b_l} \left[\frac{1}{2!} \left(\frac{1}{2} \mathbf{b}^T A^{-1} \mathbf{b} \right)^2 \right] \Big|_{\mathbf{b}=0} \\
 &= \frac{\partial}{\partial b_i} \frac{\partial}{\partial b_j} \frac{\partial}{\partial b_k} \frac{\partial}{\partial b_l} \left[\frac{1}{8} \sum_{m,n,p,q=1}^N (A^{-1})_{mn} (A^{-1})_{pq} b_m b_n b_p b_q \right] \Big|_{\mathbf{b}=0} \\
 &= \frac{1}{8} \sum_{m,n,p,q=1}^N (A^{-1})_{mn} (A^{-1})_{pq} \frac{\partial}{\partial b_i} \frac{\partial}{\partial b_j} \frac{\partial}{\partial b_k} \frac{\partial}{\partial b_l} b_m b_n b_p b_q \Big|_{\mathbf{b}=0} \\
 &= \frac{1}{8} \cdot 8 \left[(A^{-1})_{ij} (A^{-1})_{kl} + (A^{-1})_{ik} (A^{-1})_{jl} + (A^{-1})_{il} (A^{-1})_{jk} \right] \\
 &= (A^{-1})_{ij} (A^{-1})_{kl} + (A^{-1})_{ik} (A^{-1})_{jl} + (A^{-1})_{il} (A^{-1})_{jk} \\
 &= \langle x_i x_j \rangle \langle x_k x_l \rangle + \langle x_i x_k \rangle \langle x_j x_l \rangle + \langle x_i x_l \rangle \langle x_j x_k \rangle.
 \end{aligned} \tag{4.1}$$

2 Gaussian theory

Let us consider the Gaussian theory in $d < 4$ dimensions:

$$Z = \int \mathcal{D}\phi e^{-H[\phi]}, \quad (5)$$

$$H[\phi] = \int d^d r \left[\frac{1}{2} (\nabla \phi)^2 + t \phi^2 \right] \quad (t \in \mathbb{R}). \quad (6)$$

(5) Perform the Gaussian integral explicitly, and then write down the free energy density.
 [Note] You may focus solely on the most singular contribution to the free energy density and omit the t -independent constant term.

Solution. Integrating by parts, we have:

$$Z = \int \mathcal{D}\phi e^{-H[\phi]} = \int \mathcal{D}\phi \exp \left[-\frac{1}{2} \int \phi (-\nabla^2 + 2t) \phi \right] \propto [\det (-\nabla^2 + 2t)]^{-\frac{1}{2}} \quad (5.1)$$

By definition, and omitting the constant term, we obtain

$$F = -\frac{1}{\beta} \log Z = \frac{T}{2} \log \det (-\nabla^2 + 2t) \quad (5.2)$$

Since $e^{i\mathbf{k}\cdot\mathbf{r}}$ is an eigenvector of $(-\nabla^2 + 2t)$:

$$(-\nabla^2 + 2t) e^{i\mathbf{k}\cdot\mathbf{r}} = (\mathbf{k}^2 + 2t) e^{i\mathbf{k}\cdot\mathbf{r}} \quad (5.3)$$

Therefore,

$$\det (-\nabla^2 + 2t) = \prod_{\mathbf{k} \in \mathbb{R}^d} (\mathbf{k}^2 + 2t) \quad (5.4)$$

The free energy density can be expressed as:

$$f := \frac{F}{V} = \frac{T}{2V} \sum_{\mathbf{k} \in \mathbb{R}^d} \log (\mathbf{k}^2 + 2t) \stackrel{V \rightarrow \infty}{\rightsquigarrow} \frac{T}{2\mathcal{V}} \cdot \frac{\mathcal{V}}{(2\pi)^d} \int d^d \mathbf{k} \log (\mathbf{k}^2 + 2t) \quad (5.5)$$

To evaluate this integral, we first differentiate it:

$$\frac{\partial f}{\partial (2t)} = \frac{T}{2} \int \frac{d^d \mathbf{k}}{(2\pi)^d} \frac{1}{\mathbf{k}^2 + 2t} = \frac{T}{2} \frac{1}{(4\pi)^{d/2}} \Gamma \left(1 - \frac{d}{2} \right) (2t)^{d/2-1}. \quad (5.6)$$

Then, integrating both sides and dropping the constant term, we obtain the final result:

Final Answer

$$f = -\frac{T}{2} \frac{1}{(4\pi)^{d/2}} \Gamma \left(-\frac{d}{2} \right) (2t)^{d/2} \quad (5.7)$$

For odd dimensions, there is no singularity, but in even dimensions, such as $d = 2$, we need to use dimensional regularization. Using the standard expansion:

$$\Gamma \left(-1 + \frac{\epsilon}{2} \right) = - \left(\frac{2}{\epsilon} + 1 - \gamma + \mathcal{O}(\epsilon) \right), \quad \gamma = 0.577216 \dots \quad (5.8)$$

Setting $d = 2 - \epsilon$, we obtain the regularized free energy density in two dimensions:

$$f_{d=2}^{\text{ren}} = \frac{T}{2} \mu^\epsilon \left(\frac{t}{2\pi} \right)^{1-\epsilon/2} \left(\frac{2}{\epsilon} + 1 - \gamma + \mathcal{O}(\epsilon) \right) = \frac{Tt}{4\pi} \left[\frac{2}{\epsilon} + 1 - \gamma + \log(4\pi) + 2 \log \mu - \log(2t) \right] + \mathcal{O}(\epsilon) \quad (5.9)$$

Here μ^ϵ is the renormalization scale introduced to compensate for the dimension of f in $2 - \epsilon$ dimensions.

(6) For $t > 0$, calculate the most singular contribution to the specific heat c_+ and obtain the critical exponent α_+ :

$$c_+ = a_+ t^{-\alpha_+}. \quad (7)$$

Solution. We focus on the most singular term. Notice that $t = T - T_c$, therefore

$$\begin{aligned} c_+ &= -T \frac{\partial^2}{\partial T^2} f \sim \frac{T_c^2}{2} \frac{1}{(4\pi)^{d/2}} \Gamma\left(-\frac{d}{2}\right) \frac{\partial}{\partial t^2} (2t)^{d/2} \\ &= \frac{T_c^2}{2} \frac{\Gamma(2 - \frac{d}{2})}{(2\pi)^{d/2}} t^{\frac{d}{2}-2} \end{aligned} \quad (6.1)$$

The critical exponent can be read off from the equation above:

Final Answer

$$a_+ = \frac{T_c^2}{2} \frac{\Gamma(2 - \frac{d}{2})}{(2\pi)^{d/2}}, \quad \alpha_+ = \frac{4-d}{2} \quad (6.2)$$

(7) For $t < 0$, the above Gaussian path integral becomes ill defined. To regularize the theory, let us introduce a stabilizing quartic interaction and analyze Gaussian fluctuations around a stationary configuration:

$$H[\phi] = \int d^d r \left[\frac{1}{2} (\nabla \phi)^2 + t \phi^2 + u \phi^4 \right] \quad (u > 0). \quad (8)$$

On the basis of the mean-field approximation, determine the stationary point $\phi = \phi_c$ and expand the Hamiltonian $H[\phi]$ around it to the quadratic order [i.e., up to $O((\phi - \phi_c)^2)$].

Solution. We decompose the quantum field into a mean-field part ϕ_c (with $\nabla \phi = 0$) and a quantum fluctuation part η :

$$\phi(\mathbf{r}) = \phi_c + \eta(\mathbf{r}) \quad (7.1)$$

From (8), the stationary point satisfies the equation:

$$-\nabla^2 \phi + 2t\phi + 4u\phi^3 = 0 \quad (7.2)$$

For the mean field, this simplifies to:

$$2t\phi_c + 4u\phi_c^3 = 0 \Rightarrow \phi_c = 0, \quad \phi_c^2 = -\frac{t}{2u} \quad (7.3)$$

It is easy to see that, for $t < 0$, only the two solutions corresponding to the latter equation are stable stationary points, namely

$$\phi_c = \pm \sqrt{-\frac{t}{2u}} \quad (7.4)$$

Substituting this into (8) and expanding in η , we obtain:

Final Answer

$$\begin{aligned}
H[\phi_c + \eta] &= \int d^d r \left[\frac{1}{2} (\nabla \phi_c + \nabla \eta)^2 + t(\phi_c + \eta)^2 + u(\phi_c + \eta)^4 \right] \\
&= \int d^d r \left[\frac{1}{2} (\nabla \eta)^2 + \cancel{(2t\phi_c + 4u\phi_c^3)} \eta + \frac{1}{2} (2t + 12u\phi_c^2) \eta^2 + \mathcal{O}(\eta^3) \right] \\
&\quad + \int d^d r (t\phi_c^2 + u\phi_c^4) \\
&= \frac{1}{2} \int d^d r \eta (-\nabla^2 - 4t) \eta + \mathcal{O}((\phi - \phi_c)^3) + H_0
\end{aligned} \tag{7.5}$$

In the last step, we have used integration by parts.

(8) For $t < 0$, calculate the most singular contribution to the specific heat c_- and obtain the critical exponent α_- :

$$c_- = a_- (-t)^{-\alpha_-}. \tag{9}$$

Additionally, calculate the ratio a_+/a_- .

[Note] While the coefficients $a_{\pm}$ themselves are not universal, their ratio a_+/a_- is universal.

Solution. Comparing (5.1) with (7.5), we find that we only need to replace t in (6.2) by $-2t$ to obtain the free energy density under the new Hamiltonian:

$$f_- = -\frac{T}{2} \frac{1}{(4\pi)^{d/2}} \Gamma\left(-\frac{d}{2}\right) (-4t)^{d/2} \tag{8.1}$$

Following the calculation in (6.1), we obtain:

$$\begin{aligned}
c_- &= -T \frac{\partial^2}{\partial T^2} f_- \sim \frac{T_c^2}{2} \frac{1}{(4\pi)^{d/2}} \Gamma\left(-\frac{d}{2}\right) \frac{\partial}{\partial (-t)^2} (-4t)^{d/2} \\
&= \frac{T_c^2}{2} \frac{\Gamma(2 - \frac{d}{2})}{(2\pi)^{d/2}} \cdot 2^{d/2} \cdot t^{\frac{d}{2}-2}
\end{aligned} \tag{8.2}$$

Thus, we obtain the critical exponent:

Final Answer

$$\alpha_- = \frac{4-d}{2}, \quad a_- \frac{T_c^2}{2} \frac{\Gamma(2 - \frac{d}{2})}{(2\pi)^{d/2}} \cdot 2^{d/2}, \quad \frac{a_+}{a_-} = 2^{-d/2} \tag{8.3}$$